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A perspective on fluid dynamics and geochemistry coupling in geologic CO_2 storage: Key reactions, reactive transport modeling, and upscaling methods

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ABSTRACT

Understanding the complicated fluid dynamics and geochemistry coupling behaviors is the key to enhance CO_2 storage efficiency and minimize the risks of leakage and mechanical failure in geologic CO_2 storage (GCS) reservoirs. This review paper aims to discuss recent research advances associated with fluid dynamics and geochemistry coupling in GCS systems. Four research areas, i.e., flow-induced enhancement of mineral dissolution and precipitation, advanced imaging techniques based on digital core analysis techniques, advances in reactive transport modeling, and upscaling approaches from pore-scale to reservoir-scale, are covered by this review. Based on a comprehensive discussion on the research advances in the aforementioned research areas, current challenges and future research needs of fluid dynamics and geochemistry coupling in geologic CO_2 storage are highlighted. Finally, this review discusses how the research in fluid dynamics and geochemistry coupling can help in developing sustainable geologic CO_2 storage strategies that can contribute to achieving carbon neutrality goals.

1. Introduction

As fossil fuels are consumed on a massive scale, CO_2 is continuously emitted into the atmosphere, which leads to an increase in the average global temperature (IPCC, 2023; Wolf et al., 2018; Zhang et al., 2020). The increase in CO_2 concentration in the atmosphere has had a profound impact on the environment and climate of the Earth (Oelkers and Cole, 2008). CO_2 emissions are considered to be directly or indirectly linked to various environmental challenges (Goldberg and Slagle, 2009; Goldberg et al., 2008; Wolff-Boenisch et al., 2011). These include the elevation of global sea levels, frequent and intense heavy rainfall occurrences, amplified probabilities of floods and droughts, as well as the warming and ocean acidification, etc (Dessert et al., 2003; Kump et al., 2000; Oelkers and Cole, 2008). Reducing CO_2 concentrations to achieve the goal of mitigating global warming and protecting the Earth's environment has become a global consensus (Kammerer et al., 2023; Kump et al., 2000). In order to achieve this goal, it is necessary not only to control CO_2 emissions but also to implement carbon storage (Cao et al., 2023; Snæbjörnsdóttir et al., 2020; Takaya et al., 2013).

The most common method for CO_2 storage is injecting supercritical CO_2 into underground reservoirs (Jarvis et al., 2009; Li et al., 2015; Yekeen et al., 2020). Additionally, CO_2 can also be utilized to enhance the extraction of resources such as petroleum, saline water, geothermal systems, shale gas, natural gas, and others. Subsequently, CO_2 is sealed within porous rock formations beneath impermeable caprock layers (McGrail et al., 2017; Tayari and Blumsack, 2020). This will also solve the problem of coalbed methane recovery considering CO_2 sequestration in a coal seam (Prabu and Mallick, 2015), reduce peak amplitudes of nonlinear manifestations of aero-gas processes in geo-environments (Brigida et al., 2024) and overall will increase efficiency for gas from coal-shale-tight sandstone layers (Brigida et al., 2024).

Under the influence of CO_2, complex physical and chemical processes occur in rocks, which is a fluid dynamics and geochemistry coupling behavior (Brantley and Conrad, 2008; Gysi and Stefánsson, 2012; Kanakiya et al., 2017). To effectively design engineering methodologies and assess their operational efficiency and environmental repercussions, it is imperative to possess a forward-looking comprehension of fluid dynamics, the transfer of matter, hydrogeochemical transformations, and biochemical interactions within dynamic porous substrates (Golubev et al., 2005; Gysi and Stefánsson, 2012; Squires and

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Wolf, 2006). Understanding the process of fluid dynamics and geochemistry coupling is beneficial for enhancing the efficiency of geological CO_2 storage, reducing the risks of leakage, and mitigating mechanical failure risks of GCS system (Deng et al., 2022).

Over the past few decades, flow-induced enhancement of mineral dissolution and precipitation has been a crucial theme in GCS research (Awad et al., 2000; Prigiobbe et al., 2009; Wogelius and Walther, 1991). The in-situ CO_2 mineralization enhanced by flow-induced mineral dissolution and precipitation is considered a stable CO_2 sink for CO_2 sequestration (Gysi and Stefánsson, 2012; Snæbjörnsdóttir et al., 2020). Observation and comprehension of fluid dynamics and geochemistry coupling behaviors are supported by advanced imaging techniques. These imaging techniques (X-ray Computed Tomography (CT), Nuclear Magnetic Resonance (NMR), Scanning Electron Microscopy (SEM), gas adsorption, neutron imaging, etc.) reveal characteristics and changes of pore structure of geological formations under fluid dynamics and geochemistry coupling (Gan et al., 2020; Xu et al., 2019). The application of reactive transport modeling (RTM) complements laboratory imaging in microscale and supports field projects that are challenging for imaging techniques (Gostick et al., 2016; Kolditz et al., 2012; Soulaine et al., 2021; Xu et al., 2008). Process-based models in RTM are employed to forecast the movement of reacting solutes within fractures and porous substrates (Battiato et al., 2011; Deng et al., 2022; Rabbani et al., 2014). The computation of fluid dynamics and geochemistry coupling encompass different scales from pores to reservoirs, involving upscaling (Zhang et al., 2021).

A comprehensive review in this field is essential. Deng et al. (2022), Sadeghnejad et al. (2021), and Blunt et al. (2013) summarized the geochemical processes in porous media seepage during geochemical processes and reaction transport modeling methods. However, these reviews lack sufficient attention to CO_2 mineral sequestration based on fluid dynamics and geochemistry coupling and the instability factors induced by mineral reaction. Gan et al. (2020) and Xu et al. (2019) provided reviews on the development and application of advanced imaging techniques such as CT and NMR. Xu et al. (2008), Li et al. (2018), and Liu et al. (2019) conducted comprehensive reviews on the application of RTM. Zhang et al. (2021) and Yang et al. (2021) summarized the current upscaling methods. The processes of mineral dissolution and precipitation enhancement induced by fluid dynamics and geochemistry coupling (Rimstidt et al., 2012; Snæbjörnsdóttir et al., 2020), advanced imaging techniques (Blunt et al., 2013; Sadeghnejad et al., 2021), RTM (Jacquemet et al., 2009; Liu et al., 2019), and upscaling methods (Zhang et al., 2021) are four aspects with strong logical relevance and cannot be ignored when associated with GCS projects. However, there is a lack of review articles that comprehensively review and discuss these issues.

Given this, the latest research advancements in fluid dynamics and geochemistry coupling behavior within the context of GCS are summarized in this article. Based on a comprehensive discussion on the research advances in the aforementioned research areas, current challenges and future research needs of fluid dynamics and geochemistry coupling in GCS are highlighted. Finally, this review discusses how the research in fluid dynamics and geochemistry coupling can help in developing sustainable GCS strategies that can contribute to achieving carbon neutrality goals.

2. Flow-induced enhancement of mineral dissolution and precipitation

Flow-induced enhancement of mineral dissolution and precipitation occurs constantly on earth, typical examples include rock weathering (Andreani et al., 2009; Hausrath et al., 2008). Under the influence of CO_2 and water, rock minerals dissolve and secondary minerals with stable properties are formed (Gysi and Stefánsson, 2012). This geochemical process is an important step in the global carbon cycle in natural environments under multiple factors (Iglauer and Al-Yaseri, 2021), which will be introduce in this section. In-situ CO_2 mineralization storage technology and the increased risk of leakage based on enhanced mineral dissolution and precipitation will be discussed.

2.1. Geochemical processes of mineral alteration

The process of mineral transport is impacted by various factors, including chemical composition and environmental conditions (Snæbjörnsdóttir et al., 2020). Geological formations consist of various minerals, including forsterite (Hänchen et al., 2006), plagioclase (Gudbrandsson et al., 2014), pyroxene (Prigiobbe et al., 2009), basaltic glass (Gudbrandsson et al., 2008), which have relatively fast dissolution rates. The process in which CO_2 promotes the dissolution of olivine minerals in basalt, releasing metal ions and forming carbonate precipitates, is illustrated in Fig. 1 (Sandalow et al., 2021).

The release rates of ions in different minerals exhibit significant differences (Brantley and Conrad, 2008). In geological formations, forsterite is rich in Mg^{2+} and Fe^{2+}, and its dissolution rate is relatively fast (Hänchen et al., 2006). Magnesium forsterite's structure comprises interconnected silicon tetrahedra and magnesium octahedral chains (Munz et al., 2012). In this structure, magnesium-oxygen (Mg-O) bonds exhibit a quicker breakage rate compared to silicon-oxygen (Si-O) bonds. Under acid conditions, hydrogen ions catalyze the breakage of Mg-O bonds, leading to mineral degradation and dissolution (Gysi and Stefánsson, 2012). Forsterite dissolves at a faster rate compared to plagioclase (Hellevang et al., 2013). In addition to Mg^{2+} and Fe^{2+}, plagioclase can also provide a certain amount of Ca^{2+}. The dissolution rate of plagioclase exhibits a U-shaped relationship with pH (Gudbrandsson et al., 2014). In acidic environments, plagioclase dissolving rate escalates as the calcium plagioclase content rises (Gautier et al., 2001). In alkaline environments, the dissolution rate remains unaffected by the composition of plagioclase (Bachu, 2003). Basaltic glass, similar to plagioclase, has relatively slow dissolution rates despite containing various metal cations such as Mg^{2+}, Fe^{2+}, and Ca^{2+} (Cao et al., 2023). Even within the same mineral, the dissolution rates of various ions differ (Morrow et al., 2010). Studies have shown that silicon is the fastest ion released during mineral dissolution because silicon ions are released simultaneously regardless of which component dissolves (Ge, 2001). Among metal ions, magnesium ions have the fastest release rate, followed by sodium ions, iron and calcium (Gislason and Oelkers, 2003).

Mineral dissolution is significantly influenced by the ion balance of the solution, which in turn correlates with the pH value (Van Herk et al., 1989). Dissolution experiments on forsterite (Wolff-Boenisch and Galeczka, 2018), pyroxene (Kammerer et al., 2023), wollastonite (Zhang et al., 2023), and pyroxene (Diamond and Akinfiev, 2003) indicate that the main parameter controlling silicate rock dissolution is the pH of the solution rather than the carbon dioxide partial pressure (Pokrovsky and Schott, 2000). Experiments on magnesium forsterite dissolution rate, conducted in mixed-flow reactors across a broad pH spectrum from 1 to 12, reveal a consistent decrease in dissolution rate as pH increases (Oelkers and Gislason, 2001). Initially, this decrease is substantial, gradually tapering off thereafter (Kanakiya et al., 2017; Kasbohm, 2022; Liu et al., 2022; Postma et al., 2022). As pH increases, the carbonate concentration in natural water rises. This occurrence explains the reduction in the disintegration pace of magnesium forsterite as the pH level rises (Brady and Gíslason, 1997). However, there is widespread confirmation that dissolution rates decrease as pH increases under acidic conditions and increase as pH rises under alkaline conditions. Moreover, the applicability of the "U-shaped" dissolution model extends to various minerals (Coombes, 2011).

Temperature is a highly variable uncertainty factor in geological formations, varying with depth and site characteristics (Bachu, 2003). This aspect significantly affects both the breakdown of primary minerals and the creation of secondary minerals (Tester et al., 1994). Laboratory trials have investigated temperatures spanning from a few degrees

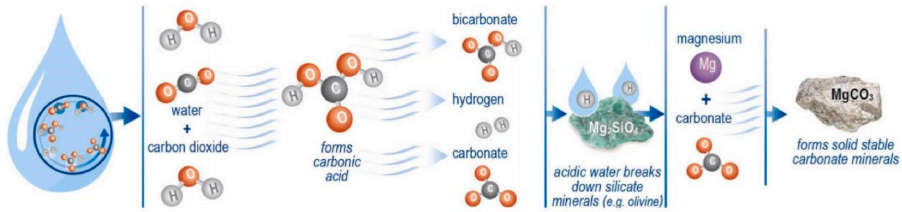

Fig. 1. Mineralization process of CO_2 with olivine in basalt formations (Sandalow et al., 2021).

Celsius to several hundred degrees Celsius. It is generally believed that the dissolution rate of minerals increases with rising temperature (Awad et al., 2000). This conclusion applies to minerals such as forsterite, plagioclase, plagioclase, pyroxene, basaltic glass, augite, quartz basalt, and serpentinite. Some studies suggest that the decrease in pH caused by increasing temperature is a factor that promotes mineral dissolution (Aftab et al., 2023; Hsieh et al., 2017; Wu et al., 2021). Contrarily, some studies suggest that as temperature rises from 100 °C to 200 °C, the liberation of Mg and Si decreases (Oelkers and Gislason, 2001). This occurrence is ascribed to the emergence of supplementary secondary silicate minerals (Carroll and Knauss, 2005).

Surface parameters are also among the primary factors influencing mineral dissolution (Yekeen et al., 2020). Surface parameters encompass various aspects such as specific surface area, surface roughness, and surface hydrophobicity. The specific surface area of minerals is positively correlated with their dissolution rate. Consequently, porous minerals exhibit faster dissolution rates compared to dense minerals, while the reaction rate of powdered minerals surpasses that of blocky minerals (Brantley and Mellott, 2000; Jeschke and Dreybrodt, 2002). However, the dynamic process of mineral reactions is complex and influenced by other factors. Although the specific surface area continues to increase, the dissolution rate of forsterite initially increases before subsequently decreasing over time (Bandstra and Brantley, 2008a). They suggest that when surface materials with etching pits dissolve rapidly, the dissolution reaction exhibits a parabolic trend.

In conclusion, the process of enhanced flow-induced mineral dissolution and precipitation is a complex interplay of multiple factors. These factors often interact and couple with each other rather than act in isolation.

2.2. In-situ CO_2 mineralization

Flow-induced enhancement of mineral dissolution and precipitation has two implications for GCS projects. Firstly, it provides a safer and faster solution for GCS, namely in-situ CO_2 mineralization sequestration (Cao et al., 2023; Takaya et al., 2013; Zhang et al., 2023), as illustrated in Fig. 2. On the other hand, this process also raise concerns about increased leakage risks, which will be discussed in the next section (Jacquemet et al., 2012; Kutchko et al., 2015).

The injection of CO_2 accelerates the dissolution rates of some basic or ultrabasic rocks (such as basalt) (Pokrovsky and Schott, 2000). The metal ions released from minerals combine with the carbonate ions generated from the dissolution of CO_2 to form chemically stable carbonate precipitates (Gislason et al., 2010). This is in-situ CO_2 mineralization (Snæbjörnsdóttir et al., 2020). Two notable engineering initiatives: The CarbFix project in Iceland involves fully combining CO_2 with water and then injecting the mixture into basalt layers. The project injected 91,294 tons of CO_2 and H_2S mixed flue gas from a geothermal power plant into basalt formations. In the initial phase of the project, the basalt formations were injected at depths ranging from 400 to 800 m, composed of basaltic lava and basaltic glass, with temperatures ranging from 20 to 33 °C. CO_2 and water were injected simultaneously into the formations at rates of 70 g/s (CO_2) and 1800 g/s (water), respectively. Monitoring showed that 95% of the injected gas was fully mineralized in situ within 2 years (Abdulelah et al., 2021). The Wallula in-situ CO_2 mineralization sequestration project near the Columbia River in Washington involved directly injecting liquid or supercritical CO_2 into basalt formations (McGrail et al., 2014; Schwartz, 2018). At a rate of 40 t/d, liquid or supercritical CO_2 was injected into basalt formations at depths of 800–900 m. The findings revealed that a significant portion of CO_2 was transformed into stable carbonate within 2 years, with monitoring showing no substantial CO_2 leakage from the reservoir. Within these cores, nodules discovered in vesicles were identified as carbonate minerals (Luhmann et al., 2017).

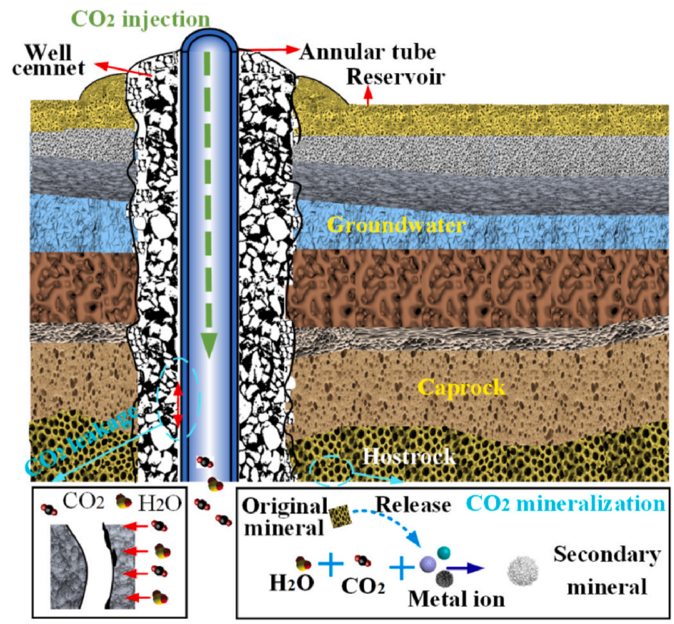

Fig. 2. Mineralization and increased leakage risk due to CO_2 injection.

2.3. Leakage risk

While flow-induced enhancement of mineral dissolution and precipitation offer benefits for GCS, on the other hand, this process may also trigger leakage risks (Beck et al., 2016; Zhang et al., 2013). After injected into saline aquifers, CO_2 alters the originally stable material structure of the formations to varying degrees (Rahman et al., 2022). This may raise concerns about engineering safety and leakage risks, although the process is often slow (Kresge and Beck, 1992; Olsen and Donald Rimstidt, 2008).

In GCS projects, achieving sealing integrity relies on casing, wellbore cement, and caprocks (Arbad et al., 2022). Regarding wellbore cement, acid gases such as CO_2 and H_2S present in GCS environments can pose a serious threat to alkaline wellbore cement (Jacquemet et al., 2012; Kutchko et al., 2015). The chemical properties and structural changes of inner and outer layers of wellbore cement after corrosion by acid gases are illustrated in Fig. 3. These acid gases dissolve the wellbore cement and generate large pores at heavily corroded regions of cement, thus compromises the structural integrity of the wellbore cement (Um et al.,

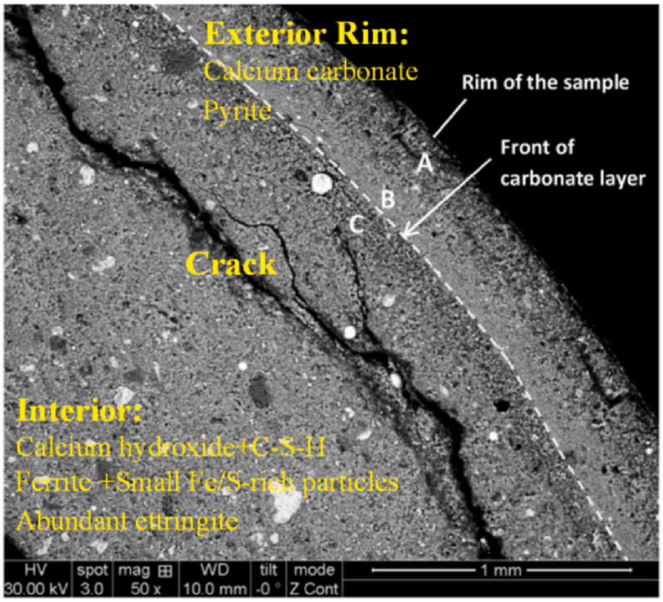

Fig. 3. Deterioration of wellbore cement caused by acid gas (Zhang et al., 2013).

2012). The degradation caused by acid gases result in increased permeability and decreased mechanical strength (Lecolier et al., 2006; Li et al., 2015; Zhang et al., 2013). Additionally, the fluid dynamics and geochemistry coupling effects can lead to caprock deformation, micro-crack propagation, increased permeability, and increased stress, which may raise the risk of CO_2 leakage. This could even activate faults and induce local seismic events (Spokas et al., 2019). Once leakage occurs, it can lead to soil contamination, groundwater pollution, and the release of heavy metals, negatively impacting the ecosystem and human health (Ye et al., 2022). Precise monitoring and numerical simulations are necessary to address these issues.

3. Advanced imaging techniques based on digital core analysis techniques

Digital core analysis techniques provide high-accuracy experimental results for GCS, including the aforementioned flow-induced enhancement of mineral dissolution and precipitation, leakage risk assessment, and other related analyses (Rani et al., 2019). Various advanced imaging techniques are used to analyze the fluid dynamics and geochemistry coupling behavior in the context of GCS, including X-ray Computed Tomography (CT), Nuclear Magnetic Resonance (NMR), Scanning Electron Microscopy (SEM), gas adsorption, and neutron imaging. The pore structure, wettability, mineral composition, evolution processes, and mechanisms of geological materials are visualized (Andrew et al., 2014; Hall et al., 2014; Um et al., 2012). The application of these techniques helps improve the quality of modeling of fluid movement in geomedia when choosing locations and parameters of CO_2 storage.

3.1. Computed Tomography (CT) scanning technology

Over the past few decades, CT scanning technology has transitioned from purely medical applications to industrial applications and has been widely used in the field of earth sciences, such as the study of porous medium pore structure and permeability characteristics (Gan et al., 2020). The advent of micro-CT has enabled CT scanning technology to achieve micron-level, and even nanometer-level, resolution (Zhang et al., 2012). Its ability to quantitatively provide detailed radiation density data has clear advantages for obtaining information on rock core frameworks (Kashim et al., 2019). In the field of GCS, CT scanning technology is used to reconstruct the distribution of rock core pore space and calculate its porosity, while also obtaining relevant parameters under fluid in-situ flow conditions, such as contact angle, relative permeability, fluid saturation, mineral generation, and distribution (Andrew et al., 2014; Garing et al., 2017; Um et al., 2012).

CT scanning technology can visualize material structures by distinguishing between the spatial distributions of pore structure and skeleton (Wang et al., 2013). The porosity and its distribution affect the fluid dynamics and geochemistry coupling process and permeability of CO_2 in reservoir rocks (Callow et al., 2018). After reconstructing images using CT scanning technology, the corresponding CT numbers can be obtained, and porosity can be calculated (Gan et al., 2020). By combining two-dimensional pore space distributions, three-dimensional pore space distributions can be obtained (Vinegar and Wellington, 1987). For dynamic processes, the expansion or sealing of channels before and after fluid flow can be observed (Lebedev et al., 2017). A typical example is the variation in rock pore distribution before and after CO_2 injection shown in Fig. 4 by Kashim et al. (2019).

CT scanning technology, as an effective method for identifying in-situ multiphase flow, can provide fluid distribution images with good spatial resolution (Turner et al., 2004). CT scanning is rapid and non-invasive, allowing observation of multiphase flow by identifying fluid saturation (Akin, 2003). Because of resolution constraints, saturation profiles may diverge from true values, and fluid migration in extremely small pores may be indistinguishable (Zhang et al., 2012). CT scanning technology can ascertain material parameters (Andrew et al., 2014; Garing et al., 2017). CT scanning technology enables the observation of chemical processes (Soong et al., 2016). However, minerals with X-ray attenuation values too close cannot be distinguished, it is necessary to complement it with other mineral component analysis tools (Reyes et al., 2017).

3.2. Other imaging techniques

In addition to CT scanning technology, advanced imaging techniques currently applied in GCS include Nuclear Magnetic Resonance (NMR) (Li et al., 2018). NMR utilizes a strong magnetic field and radio frequency pulses to excite hydrogen nuclei to produce resonance signals, which then gradually decay. NMR measures different longitudinal relaxation times T1 (reflecting the time for the nuclear spin system to return to equilibrium) and transverse relaxation times T2 (reflecting the time for the loss of nuclear spin coherence) of samples in different environments, providing information about pore structure and fluid properties (Zhao et al., 2013). In digital core analysis, NMR can be used to determine core

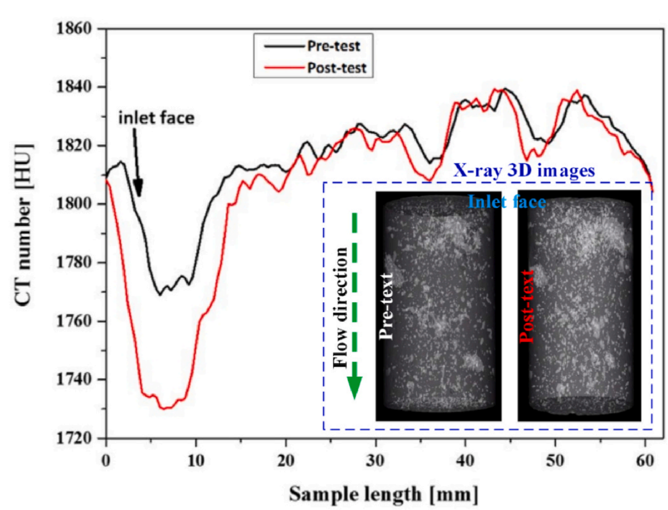

Fig. 4. Porosity and structural changes with the use of a CT scanner (Kashim et al., 2019).

porosity and pore size distribution, measure core permeability, and monitor the distribution and migration of fluids within the core (Mohnke and Yaramanci, 2002; Teng et al., 2017; Zhang et al., 2015). A typical case involves using NMR to observe the effects of different CO_2 injection schemes on oil recovery in ultra-low permeability sandstone, with the changes in the T2 spectra of the samples shown in Fig. 5 (Xiao et al., 2017). Similar to other non-destructive testing techniques, NMR also has advantages such as being non-destructive, non-toxic, harmless, pollution-free, rapid, and simple (Clarke et al., 1995). Additionally, while CT technology cannot reflect interactions between fluids or between fluids and the solid matrix, NMR is sensitive to fluids, giving it certain advantages over CT (Liu et al., 2015). However, there are also challenges when applying NMR to GCS, such as the development of high-precision pore models and permeability empirical formulas, the acceleration of signal decay due to the paramagnetic nature of rocks (Bencsik and Ramanathan, 2001), the lack of commercial NMR post-processing software for formation materials (Wang et al., 2022), and signal interference issues in multi-parameter monitoring (Xu et al., 2019), etc.

SEM-EDS is commonly used for micro-scale morphological observation and sample analysis of various engineering geological specimens such as minerals, rocks, and cement materials (Loucks et al., 2009). The imaging resolution of SEM can reach the nanometer level. SEM and EDS are both surface analysis devices based on the interaction of electron beams with matter (Yekeen et al., 2020). They allow for direct observation of the fine structures of various samples' uneven surfaces (Brantley and Mellott, 2000; Jeschke and Dreybrodt, 2002). Scanning of sample thin sections using scanning electron microscopy enables convenient acquisition of space data. The use of SEM and EDS also allows for detailed mineralogical characterization, which can complement the morphological data from CT images (Feng and Gray, 2017; Li et al., 2023). Scanning electron microscopy is proficient in generating two-dimensional images of microstructures but lacks the capability to provide three-dimensional images (Gharasoo et al., 2012). However, three-dimensional images are crucial for understanding interconnected pores in three-dimensional space and evaluating the three-dimensional heterogeneity of pore structures (Gan et al., 2020). Gas adsorption is an accurate technique for determining pore size, but it can only measure open pore volume and cannot measure isolated pore volume (Rani et al., 2019). Neutron imaging is limited by the high equipment costs associated with its implementation (Hall et al., 2014).

4. Advances in reactive transport modeling

Digital core analysis techniques are precise. However, existing imaging techniques are costly, make large-scale in-situ observations challenging, and have short observation time scales. Reactive transport modeling provide valuable tools for analysis from the pore scale to the reservoir scale (Kolditz et al., 2012; Noiriel and Soulaine, 2021; Permann et al., 2020). In GCS systems, the interaction of CO_2-water-rock is a complex process involving multiple physical, chemical, and biological processes forming a multi-coupling system (Black et al., 2015). Fluid dynamics and geochemistry coupling feeds back into flow, transport, and chemical reactions (Kumar et al., 2020). The method of RTM for complex processes is considered efficient in GCS system (Xu et al., 2011). RTM in GCS inherit the theoretical frameworks of all relevant independent disciplines and introduce additional methodologies to couple these processes (White et al., 2005).

The foundational theory of RTM includes energy conservation, fluid mass conservation, solute mass conservation, linear or nonlinear laws applicable to continuous porous media flow, laws governing mass interactions between liquid/gas phase components and equilibrium mineral phases, kinetic rate equations for components and minerals, as well as fundamental laws of various major processes (Audigane et al., 2007; Geloni et al., 2011; Zhang et al., 2009). These theories do not exist in isolation but interact with each other, such as coupling between energy and momentum conservation, including thermal convection and temperature effects on fluid density and viscosity induced by fluid flow (Gunter et al., 1993); coupling between fluid motion and solute concentration, including solute convection and concentration effects on fluid density and basic properties of porous media induced by fluid flow; coupling between energy conservation and solute concentration, this encompassed accounting for the impacts of temperature on thermodynamics and reaction rates, as well as considering the influence of chemical reactions on heat. (Alkan et al., 2010). In addition to the above coupling systems, there are also couplings of multiple complex systems (Thompson et al., 2019). Coupled systems are nonlinear systems whose complexity depends on the complexity of subsystems and the number of coupled subsystems (Liu et al., 2011).

Efficient and accurate codes are valuable for RTM involving fluid dynamics and geochemistry coupling behavior (Izgec et al., 2008). Currently, there are many well-validated codes and publicly available experimental and simulation data (Lei et al., 2012). Some of the codes currently applied are summarized in Table 1. Among them, the TOUGH series software, developed by the Lawrence Berkeley National Laboratory in the United States, can handle various issues such as geothermal energy development, unsaturated zone water and gas migration, oil and gas migration, GCS, natural gas hydrate development, and environmental remediation. TOUGHREACT, based on TOUGH2, adds a multi-phase flow solute transport module, providing advantages in addressing complex water-rock interactions (Xu et al., 2008). PHREEQC is a geochemical reaction calculation software based on the ion association water model, capable of simulating geochemical inverse processes and one-dimensional transport reactions in GCS. OpenGeoSys is a simulator developed by the Helmholtz Centre in Germany based on FEM, capable of simulating single or coupled THMC processes in porous and fractured media (Kolditz et al., 2012). In GCS simulations, other available codes or programs include MIN3P, MIN3P (Sihota and Mayer, 2012), CrunchFlow (Steefel et al., 2015), Pflotran (Steefel et al., 2015), Mgstat (Hansen, 2004), OpenPNM (Gostick et al., 2016),porousMedia4Foam (Soulaine et al., 2021), geochemFoam (Maes and Menke, 2021), PNBRNS (Gharasoo et al., 2012), SGeMS (Remy, 2005), STOMP (White et al., 2020), MOOSE (Permann et al., 2020).

Next, several applications of reactive transport modeling and code in GCS systems are illustrated: CO_2-water-formation interaction, CO_2 leakage risk prediction, and assessment of the environmental impact of CO_2 leakage.

4.1. RTM applications in GCS: CO_2-water-rock interaction

The storage amount and mineralization process of CO_2 in specific

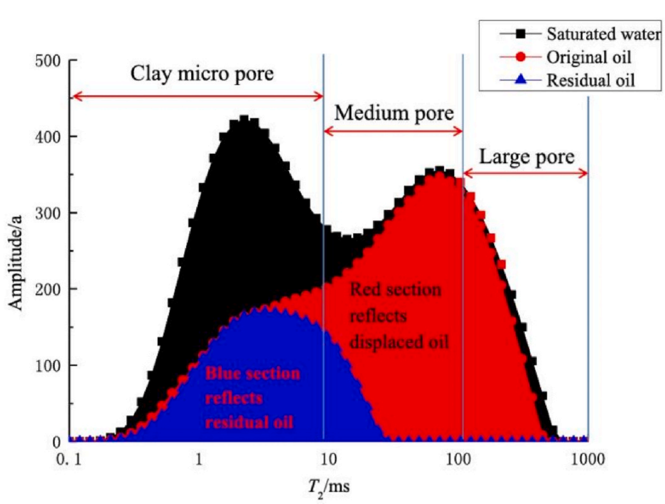

Fig. 5. T2 spectrum (Xiao et al., 2017).

Table 1
Comparative table for the current available algorithms/codes.

Name	Applicability range	References
TOUGH series codes	A suite of simulators for multiphase fluid and heat flow in porous and fractured media, widely used in geothermal reservoir engineering, nuclear waste disposal, and GCS.	Xu et al. (2008)
CrunchFlow	A code for modeling reactive transport in geochemical systems, simulating fluid flow and the interaction of chemical species in subsurface environments.	(Steefel et al., 2015; Zhang et al., 2017)
OpenGeoSys	An open-source initiative for numerical simulation of thermo-hydro-mechanical/chemical (THM/C) processes, used in geological and environmental engineering for multiphysical problems.	Benisch et al. (2013)
CMG	Computer Modeling Group offers advanced software solutions for reservoir simulation, supporting multiphase fluid flow and complex reactive transport processes.	Ginting et al. (2019)
PHREEQC	A versatile geochemical reaction modeling program widely used for water chemistry, pollutant transport, and mineral dissolution/precipitation studies.	Pötter et al. (2021)
MIN3P	A three-dimensional multiphase and multicomponent reactive transport model used for studying groundwater and soil contamination.	Bao et al. (2017)
Pflotran	An open-source parallel code for large-scale subsurface flow and reactive transport simulations, supporting multiphysics and multiphase processes.	Orsini et al. (2014)
Mgstat	A tool for geophysical modeling, supporting uncertainty analysis and inversion of geological properties.	Hansen (2004)
geochemFoam	A geochemical reactive transport simulation tool based on OpenFOAM, combining capabilities for fluid dynamics and geochemical reactions.	Orsini et al. (2014)
MOOSE	Multiphysics Object Oriented Simulation Environment, a flexible platform for simulating multiphysics coupling.	Permann et al. (2020)
STOMP	Subsurface Transport Over Multiple Phases, a code for simulating multiphase and multicomponent fluid flow and reactive transport, widely used in environmental engineering and resource management.	Nguyen et al. (2016)
SGeMS	Stanford Geostatistical Modeling Software, used for modeling and simulation of geological properties.	Remy (2005)
COMSOL	A multiphysics simulation software that provides solutions for modeling and simulating various physical and chemical processes, including fluid flow, heat transfer, and reactive transport, widely used in scientific research.	Azad et al. (2016)

formations are essential considerations, and RTM provides efficient analytical tools (Tempel and Harrison, 2000). Xu et al. (2004) utilized the reactive transport simulator TOUGHREACT to assess the storing capacity in three formations. The capacities were 17 kg/m^3 for the Green River Sandstone aquifer, 90 kg/m^3 for the Gulf Coast aquifer, and 100 kg/m^3 for the Dunite rock formation. CO_2 injection led to reduced porosity, affecting the carbon storage capacity of the solid matrix (Xu et al., 2004). White et al. (2005) numerically assessed the CO_2 storage potential at a site in the Colorado Plateau of central Utah and modeled CO_2 injection into the sandstone using ChemTOUGH. According to the findings, approximately 21% of the CO_2 was sequestered as minerals after 1000 years, while 52% dissolved in water as gas, and 17% leaked. Kumar et al. (2020) simulated CO_2 storage volumes and mineralization rates in the limestone aquifer of the South Florida Basin under different temperatures using TOUGHREACT, showing a 35% reduction in storage efficiency as temperatures increased from 35 °C to 95 °C. CO_2 dissolution trapping increased with temperature. For further understanding of CO_2 storage and evolution processes in different formations, studies by Audigane et al. (2007), Zhang et al. (2009), and others have been conducted. Calculations conducted by Whang et al., (2016) illustrate the evolution of the injected CO_2 within the geological formation as depicted in Fig. 6. Among these, the percentage of CO_2 sequestration as mineral shows a stable increase after 10 years of injection, which is considered a stable sequestration mode.

The trapping mechanisms of injected CO_2 in formations involve complex fluid dynamics and geochemistry coupling process (Knauss et al., 2003). Gunter et al. (1993) investigated CO_2 dissolution mechanisms, indicating unfavorable for absorption of brine and CO_2 reactions in carbonate aquifers involving dissolution of calcite. Dissolved CO_2 in siliciclastic aquifers neutralizes through reactions between plagioclase and kaolinite, significantly enhancing the aquifer's capacity to handle CO_2. Liu et al. (2012) conducted a reaction-transport simulation to analyze the long-term evolution (over 100 years) of CO_2 injection (1 million tons/year) into the Mount Simon Sandstone in the United States. They assessed four primary trapping mechanisms (hydrodynamic, solubility, residual, and mineral trapping) along with their spatiotemporal variations. Izgec et al. (2008) analyzed factors affecting CO_2 storage efficiency over short simulation periods, highlighting the significance of fluid dissolution compared to mineral trapping and adsorption.

The total CO_2 storage and efficiency in formations are influenced by various factors such as CO_2 partial pressure, temperature, rock porosity, and brine salinity. Tempel and Harrison (2000) evaluated the impact of different CO_2 partial pressures, fluid compositions, rock reactive surface areas, and burial histories on Wilcox sandstone diagenesis, showing that CO_2 interaction with Wilcox sandstone promoted carbonate precipitation and inhibited secondary porosity formation, consistent with observations by Fisher and Land (1987) on Wilcox diagenesis. Alkan et al. (2010) discussed the effects of capillary pressure, salinity, and field conditions on CO_2-water-rock interactions, revealing outward diffusion of free CO_2 below the caprock, filling rock pores with CO_2 gas within about 60 m radially within 10 years. Kumar et al. (2020) focused on the influence of formation temperature. Zhao et al. (2019), Liu et al. (2015), Tang et al. (2019), and Vermolen et al. (2009) elucidated the effects of flow model parameters on two-phase non-miscible displacement under fluid dynamics and geochemistry coupling. Notably, some impurity gases also significantly impact CO_2-formation interactions (Vermolen et al., 2009). Knauss et al. (2003), Spycher et al. (2019), among others, conducted reactive transport simulations to study the effects of dissolved CO_2, H_2S, N_2, SO_2 mixtures on formations. This is a noteworthy issue but has not yet received sufficient attention.

The modification of native formation rocks resulting from CO_2 injection can alter the chemical composition and pore structure of formations. Ennis-King and Paterson (2007), Lindeberg and Wessel-Berg (1997), and others simulated the process of dissolved CO_2 ions such as Na^+, Ca^{2+}, Mg^{2+}, and Fe^{2+} complexing to form new minerals over time (Ennis-King and Paterson, 2007; Lindeberg and Wessel-Berg, 1997). The impact of CO_2-water-rock reactions on surface features of rocks, as well as the alteration of pore structure, is crucial and can also be simulated using reactive transport models (Brantley and Mellott, 2000; Jeschke and Dreybrodt, 2002; Bandstra and Brantley, 2008a, b). Dashtian et al. (2019) analyzed the advection, diffusion, and reaction mechanisms of water-CO_2-rock through simulation, predicting changes in structure and rock physical properties using reactive solute transport and dissolution (Dashtian et al., 2019). CO_2 mineral trapping significantly influences changes in pore structure morphology. While some reactive transport simulation results show an increase in primary mineral porosity due to

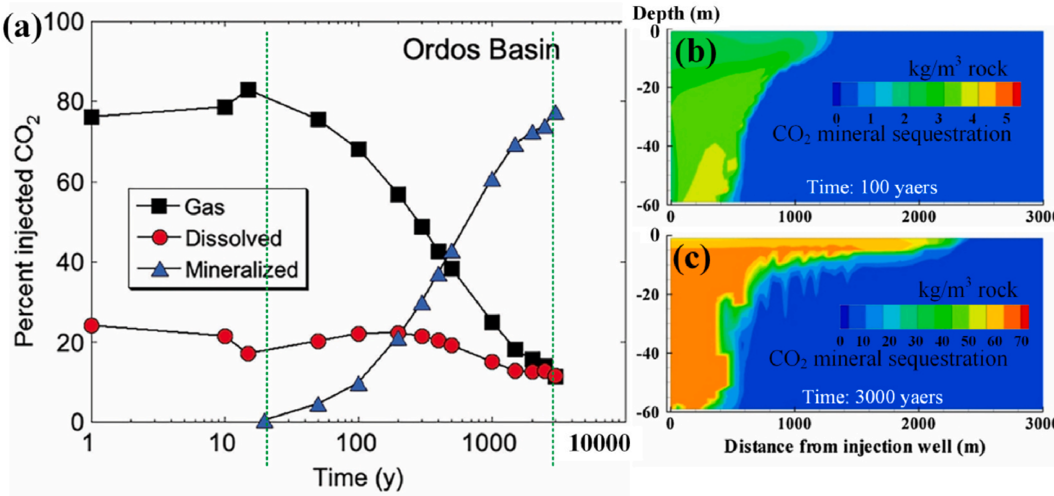

Fig. 6. The evolution of the injected CO_2 in the formation: (a)Time evolution in mass fractions of the injected CO_2 in different trapping mechanisms; (b)spatial distribution of CO_2 as mineral sequestration after 100 years of injection; (c) spatial distribution of CO_2 as mineral sequestration after 3000 years of injection (Wang et al., 2016).

CO_2-water-rock reactions, considerable research suggests that secondary carbonate formation may lead to pore space reduction (Xiao et al., 2017). More specific and detailed data are still needed to assess the behavior and impacts of CO_2 storage in formations using reactive transport models at specific locations and to aid in the engineering design of GCS systems (Fisher and Land, 1987).

4.2. RTM applications in GCS (2): assess the possibility and path of CO_2 leakage

CO_2 injection can cause alterations in the geological formations, potentially impacting the sealing integrity of GCS systems (Johnson et al., 2005). Quantitatively assessing the likelihood of leaks in GCS systems is another application scenario for RTM (Gherardi et al., 2007; Soltanian1 et al., 1969). Johnson et al. (2005) utilized RTM to assess the evolution of cap rock permeability and the long-term hydrodynamic sealing capacity of storage sites after CO_2 sequestration in target formations. Gherardi et al. (2007) conducted numerical simulations on the leakage and migration of CO_2 in cap rocks of onshore depleted gas reservoirs due to GCS. The simulation results indicated that CO_2 migration through fractures to the cap rock would decrease the pH value, significantly dissolve calcite in the cap rock, increase porosity significantly, and when the porosity water content is high, calcite would re-form and reseal the cap rock pores. Additionally, the process of CO_2 corroding wellbores and causing leaks can also be predicted through RTM. For instance, Geloni et al. (2011) used the TOUGHREACT simulator with the TMGAS EOS module to study the impact of CO_2 injection on the structural integrity of rock-cement structures. The model was capable of predicting mineralogical alterations happening at the interfaces between the cement and reservoir, as well as between the cement and cap rock. It also simulated the processes of muscovite dissolution, as well as the precipitation of zeolites and brucite (Geloni et al., 2011). Such simulations can provide preliminary data for CO_2 storage site selection (Gherardi et al., 2007).

4.3. RTM applications in GCS (3): impact of CO_2 leakage on the environment

The alteration process of cap rock minerals exhibits uncertainty. Considering the potential environmental impacts following a leak is also necessary (Soltanian1 et al., 1969). Reactive transport models can not only elucidate the spatial and temporal scales of fluid dynamics and geochemistry coupling processes but also predict the potential consequences of leakage (Alkan et al., 2010; Zheng et al., 2010). CO_2 leakage may lead to contamination of shallow aquifers, release of heavy metal pollutants, and even pose risks to human survival on the surface (Birkholzer et al., 2008). Many scholars have attempted to assess these consequences using reactive transport models. Wang and Jaffe (2004) developed a RTM by considering the aquifer to comprise solely quartz, calcite, and galena, simplifying the geochemical system to assess the negative impacts of CO_2 leakage on groundwater. The findings revealed that CO_2 leakage resulted in a decline in groundwater pH, substantial dissolution of galena, and an elevation in lead ion concentration in the water. Zheng et al. (2010) employed a continuum-scale RTM to study the release of shallow heavy metals caused by CO_2 and saline water leaking into shallow aquifers. Carroll et al. (2005) evaluated the impact of CO_2 leakage on the pH of water in the US High Plains aquifer through reactive transport simulations, without considering changes in trace element concentrations. Similar studies were conducted by Birkholzer et al. (2008), who incorporated thermodynamic analysis into reactive transport models. The research indicated that CO_2 leakage into shallow aquifers could result in the release of harmful trace elements such as As and Pb. Simulation results by Alkan et al. (2010) demonstrated that CO_2 injected beneath reservoirs flowed and formed a complete drying zone over a period of 10 years, extending vertically to the top of the reservoir. Without the assistance of RTM, this complex fluid dynamics and geochemistry coupling process would be unpredictable (Wang and Jaffe, 2004).

5. Upscaling approaches from pore-scale to reservoir-scale

As can be seen in the above discussion, the fluid dynamics and geochemistry coupling processes in GCS systems involves multiple scales, making upscaling a critical issue (Moslehi et al., 2016). In recent years, upscaling has garnered considerable attention as a prominent challenge in GCS projects (Deng et al., 2022; Zhang et al., 2021). This complex process involves research spanning from the pore-scale to the reservoir-scale and encompasses various geochemical systems and dynamics, as illustrated in Fig. 7. Fluid dynamics and geochemistry coupling behaviors in GCS systems at different scales exhibit strong heterogeneity, and studying their distribution and upscaling transport is complex (Noiriel and Daval, 2017). Heterogeneity is pervasive at all scales, including mineral composition, stratification, lithological differences, fracture distribution, mineral shape, porosity, pore geometry, and their distributions, among other physical-chemical properties (Zhao et al., 2019).

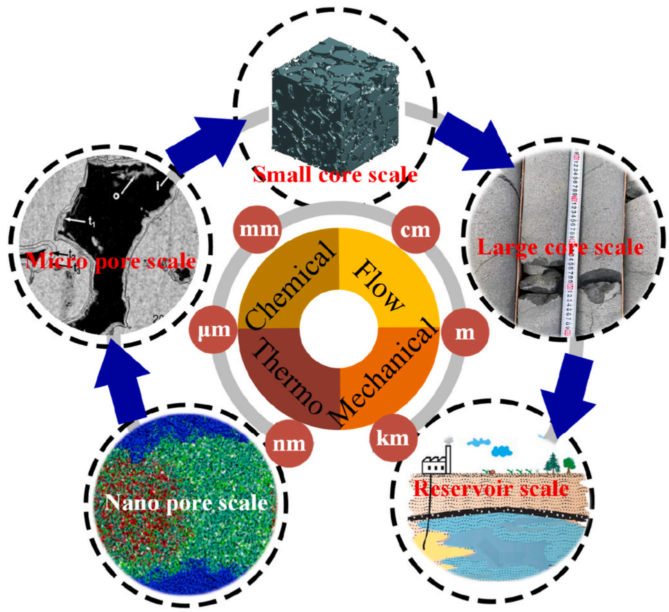

Fig. 7. Upscaling approaches from nano-scale to reservoir-scale (adapted from (Deng et al., 2022; Noiriel et al., 2007)).

Generally, research is precise under pore scale (Yang et al., 2021). Within the pore-scale domain, characteristics including chemical changes, structural alterations of solids, pores, and fluids can be observed or calculated (Zacharoudiou et al., 2018). The application and development of precise observation methods have facilitated the understanding of fluid dynamics and geochemistry coupling behavior at the pore scale (detailed descriptions of these imaging techniques will be provided in the following sections of this paper) (Cecen et al., 2018). For example, Tang et al. (2019) and Zacharoudiou et al. (2018) simulated CO_2 displacement on 3D CT images. Fig. 8 illustrates the process of obtaining the real morphology of pore throats from 3D CT data of Berea sandstone and then extracting the sample network model (Rabbani et al., 2014). Direct Numerical Simulation (DNS) entails directly solving the governing equations of the pore space image (Soulaine et al., 2021), employing methods such as Level-Set (LS) (Prodanović et al., 2010) and Volume-of-Fluid (VoF) methods (Raeini et al., 2017). Additionally, methods such as the pore network model (PNM) and the lattice Boltzmann grid model are utilized to approximate solutions for the Navier-Stokes equations (Acharya et al., 2005).

Pore-scale models can typically distinguish between solid and pore phases and accurately describe fluid dynamics and geochemistry coupling processes at the solid-liquid interface (Blunt et al., 2013; Hajizadeh and Farhadpour, 2012). Nevertheless, pore-scale analysis is constrained by relatively small computational domains. Precise analysis at the pore scale comes with expensive analysis costs (Zhang et al., 2021). Hybrid models and micro-continuum models (MCM) have been developed. The idea of hybrid models is to divide the model into multiple subdomains, apply different model parameters, control equations, and boundary conditions to the decomposed subdomains, and integrate the analyzed subdomains. Hybrid models can handle local heterogeneity (Battiato et al., 2011). MCM distinguish between pore-scale domains and porous domains based on local porosity (p) (Noiriel and Soulaine, 2021). Flow is governed by Darcy-Brinkman-Stokes equation, which is applicable when the solid phase exhibits microporous or nanoporous characteristics. The entire computational domain of micro-continuum models solves only one set of governing equations (Zhang et al., 2022).

At larger scales, to reduce computational costs, methods have been developed to replace non-uniform domains with uniform ones (Lichtner and Kang, 2007). Continuous-scale analysis involves averaging over REV (Bear, 1972). Microscopic information is used to simplify calculations (Xu et al., 2004; Yang et al., 2021). The analysis requires introduction of some representative parameters, such as homogeneous equations, and major constitutive relationships (Noiriel and Soulaine, 2021; Zhang et al., 2022). Although this process simplifies calculations, it evidently overlooks the influence of reservoir heterogeneity (Hajizadeh and Farhadpour, 2012).

The ability to integrate analysis of fluid dynamics and geochemistry coupling processes at different scales determines the safety, accuracy, and economy of GCS engineering (Ellis et al., 2013; Mangane et al., 2013; Zhao et al., 2019). For accurate and effective predictive modeling, the development of multiscale strategies is essential. These strategies should capture crucial processes with varying characteristic lengths and time scales, while also transitioning to finer scales through high-fidelity models. Heterogeneity is a difficulty in upscaling research (Xu et al., 2004). The forthcoming section will begin by examining the scale dependency of GCS systems across multiple parameters. Subsequently, theories pertaining to upscaling will be summarized.

5.1. Scale dependency of GCS systems

5.1.1. Mineral dissolution rates

The dissolution rate of mineral exhibits scale dependency (Maher and Chamberlain, 2014). Section 2 provided a detailed description of mineral dissolution, accompanied by laboratory experiments conducted across scales ranging from millimeters to meters (Morse and Arvidson, 2002; Vriens et al., 2020). In addition to laboratory experiments, field experiments are also widely conducted (Sverdrup, 2009). At the field scale, mineral dissolution rates are generally assessed by observing mineral depletion fronts or ion concentrations in water (Jung and Navarre-Sitchler, 2018). Determination of ion concentrations in water flow can be carried out on scales ranging from meters to kilometers (De Windt et al., 2004). Mineral dissolution rates exhibit strong mineral differences as well as scale differences. Mineral dissolution rates determined at the field scale may be 4-5 orders of magnitude lower than laboratory measurements under similar conditions, as reported by White et al. (1996) regarding plagioclase dissolution rates in the field. When other conditions are the same, the measured mineral dissolution rate decreases as the flow path lengthens. Besides exhibiting scale

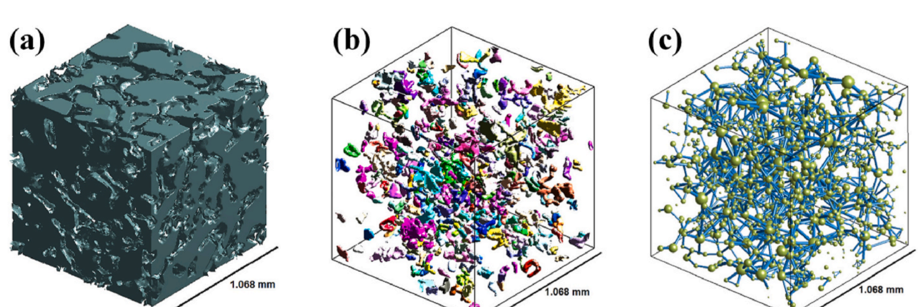

Fig. 8. (a) CT 3D images; (b) Real shape of throats; (c) The network model of Berea sandstone (Rabbani et al., 2014).

dependency spatially, mineral dissolution rates also demonstrate complexity over time scales (Moore et al., 2012). Within the fluid dynamics and geochemistry coupling process in GCS systems, the presence of heterogeneity adds complexity to the changes in mineral dissolution across spatial and temporal dimensions (Gysi and Stefánsson, 2012).

5.1.2. Transport parameters

Experimental and numerical simulation results both indicate that dispersivity exhibits scale effects (Dagan, 1984). The range of dispersivity variation typically spans over 2-3 orders of magnitude. Previous studies on porous and fractured media dispersivity suggest that longitudinal dispersivity increases with scale (Appelo and Postma, 2004). Dispersion depends on the velocity field and increases with the gradient of the longitudinal velocity field (Klotz et al., 1980). Hence, lateral diffusion of solute transport is limited. Additionally, inaccuracies in porosity measurements can lead to errors in dispersivity measurements (Abgaze and Sharma, 2015a).

In upscaling analysis, diffusion coefficient also exhibit strong scale dependency (Liu et al., 2004). Diffusion coefficients at field scales (measured in kilometers) often surpass those observed at laboratory scales and typically increase with the enlargement of the test scale (Reimus and Callahan, 2007). In crystalline rocks, diffusion coefficients at kilometer scales exceed those measured in the laboratory by at least three orders of magnitude (Neretnieks, 2002). Minor fractures can significantly increase the surface area of the fracture-matrix interface, a characteristic that numerical models often fail to adequately represent. This limitation might explain the relatively high effective diffusion coefficients observed in field data (Rolle et al., 2013).

Sorption plays a crucial role in decelerating CO_2 migration within aquifers (Zhang et al., 2014). The heterogeneity of reservoir materials will significantly influence the assessment of sorption (Limousin et al., 2007). CO_2 is transported through aquifers via groundwater, and their migration rates may be slower than the average groundwater flow velocity (Dai et al., 2020). The adsorption rates of radionuclides in fractured crystalline rocks are significantly higher by 3–4 orders of magnitude compared to the results observed at the field scale.

5.1.3. Hydraulic conductivity tensor

The hydraulic conductivity tensor determines the flow field state in the formation. Existing experimental results consistently indicate significant variations in hydraulic conductivity (K) with scale (Chapuis et al., 2005). These studies compare laboratory experiments at relatively small scales with medium to large-scale aquifer tests. The characteristic scale-dependent variation of K is mainly observed in heterogeneous materials, while in relatively homogeneous materials, the value of K remains relatively constant (Abgaze and Sharma, 2015b). Geological structures such as unconsolidated sediments and fractured rock formations, which are prevalent in GCS systems, exhibit significant scale effects. Studies indicate that the value of K increases with the expansion of scale (Azizmohammadi and Matthäi, 2017). Statistical results from Zhang et al. (2022) demonstrate a wide range of K values (varying by several orders of magnitude) at a constant scale. As the characterized volume exceeds 1 m^3, with an expanding test radius, there is a minor increase observed in K. At the microscopic scale, there is no significant variation in K with volume.

5.2. Theories for upscaling

5.2.1. The upscaling methods for hydraulic conductivity

Upscaling methods for hydraulic conductivity commonly used in upscaling include spatial averaging, cross-scale fractal geometry, wavelet transformation, and image machine learning (Zhang et al., 2021). The hydraulic conductivity at larger scales can be estimated by averaging the values of each block in microscale grid units (calculated using power averaging methods) (Bandstra and Brantley, 2008b; Deng et al., 2022). Indeed, globally upscaling presents computational hurdles since addressing the entire domain at fine scales may become impractical due to excessive memory demands (Xu et al., 2004). In practical applications of GCS engineering, the majority of aquifers are assumed to be isotropic to mitigate the computational expense linked with upscaling (Ellis et al., 2013; Noiriel and Soulaine, 2021; Yang et al., 2021). The error stemming from assuming isotropy is relatively minimal compared to the overall error introduced by scaling and has been verified (Scheibe and Yabusaki, 1998).

Fractal geometry was introduced into hydrogeology to elucidate the irregular flow-chemistry pathways. This was accomplished by quantifying the fractal dimension of groundwater flow and transport (Ghanbarian and Hunt, 2017). Fractals refer to geometric objects with scale invariance (White et al., 1996). The influence of scale-up is greatly reduced in fractal geometry. GCS systems consist of porous media or fractured media, whose pores or fractures can be viewed as fractals (Guéguen et al., 2006).

5.2.2. Multi-parameter upscaling method

The conceptual model of upscaling entails the transformation of "local" parameters from a fine scale to "block" parameters on a coarse scale is shown in Fig. 9. Currently, methods for multi-parameter upscaling mainly include homogenization, moment and volume averaging, etc. (Zhang et al., 2021)

The upscaling of heterogeneity in GCS systems can be addressed through homogenization (Allaire, 1992). During the upscaling process, the heterogeneous porous media structure is transformed into a periodic structure (Diwakar and Kumar, 2018). Homogenization is achieved by seeking average formulas through asymptotic analysis to describe periodic arrays. The homogenization theory is relevant for partial differential equations characterized by coefficients that fluctuate rapidly across multiple scales (Borges et al., 2007). In periodic media, the period's size is considerably smaller than the medium's sample size (Xu et al., 2004). Homogenization theory has been effectively employed in addressing diverse groundwater flow and solute transport issues (Chou et al., 1989).

The method of moments is a statistical method used to estimate population parameters (Dill and Brenner, 1980). If the estimated value of a population parameter matches the moment of the sample, then the sample statistic can be estimated, and the population moment can be substituted into the equation (Aris, 1956). This method can be used to estimate effective permeability of blocks through local permeability estimation (White et al., 1996). Unlike estimating block permeability through average flow rates and hydraulic gradients (Adrover et al., 2019), the values obtained by the method of moments correspond to spatial moments of hydraulic heads in heterogeneous media (Chrysikopoulos et al., 1992).

The fundamental concept of the volume averaging method is to upscale the solute transport control equation from Darcy's scale within a closed volume (Porta et al., 2012). The heterogeneous reservoir is considered as a composite of multiple homogeneous components, encompassing various regions of porosity and solids (Bandstra and Brantley, 2008b; Deng et al., 2022), different mineral types, and diverse fluid and biofilm phases (Schmid et al., 2019). Statistically, the formation remains stationary without significant alterations (Dai et al., 2020). This occurs because, upon the return of the averaged volume to the reservoir, the volume fractions of each phase within the averaged volume remain constant (Dagan, 1984). Through the application of the volume averaging operator to the advection and dispersion terms in the transport equations of multi-region groundwater flow and transport, it is possible to derive the volume-averaged equation for transport in porous media with mineral surfaces (Mayerich et al., 2022). To ensure successful smoothing of variables in space, it is crucial to carefully select the shape and size of the appropriate averaging volume (Zhang et al., 2021). Dai et al. (2020) upscaled diffusion coefficients of multi-mode fractured heterogeneous rocks from laboratory scale to field scale.

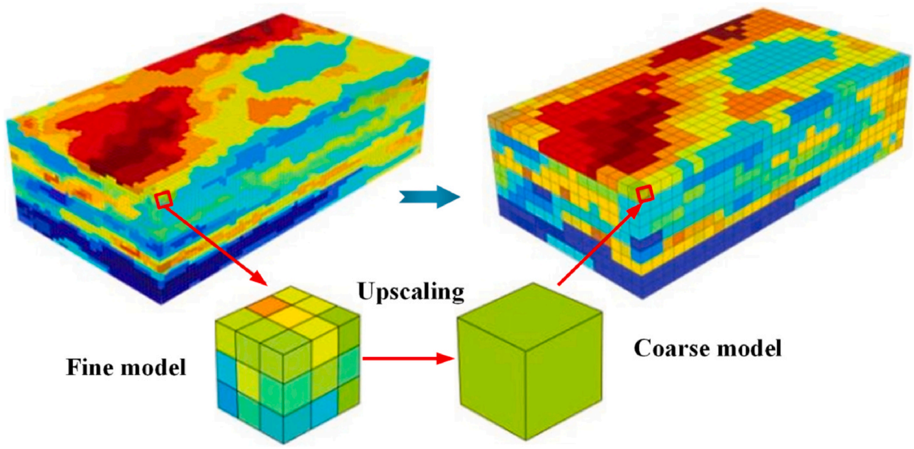

Fig. 9. Conceptual model of upscaling "local" parameters on a fine scale to "block" parameters on a coarse scale (Zhang et al., 2021).

6. Discussion

In the preceding chapters, the latest research progress in the fluid dynamics and geochemistry coupling within GCS systems has been summarized. Building upon the review of the research progress in these areas, this section discusses the challenges and future research directions of fluid dynamics and geochemistry coupling behavior in GCS. These discussions encompass: impact of CO_2 on the properties of basalt under the scenario of in-situ CO_2 mineralization in basalt, limitations on resolution and post-processing methods of imaging techniques, and influence of complex rock and gas compositions on the accuracy of RTM.

6.1. Challenges

6.1.1. Impact of CO_2 on the properties of basalt under the scenario of in-situ CO_2 mineralization in basalt

In the preceding sections, the process of flow-induced enhancement of mineral dissolution and precipitation in GCS systems has been summarized. It is understood that fluid dynamics and geochemistry coupling behaviors can alter the chemical and physical properties of formation materials (Kresge and Beck, 1992; Olsen and Donald Rimstidt, 2008). This process not only provides new insights into in-situ CO_2 mineralization but also raises concerns about the stability of basalt formations in GCS system.

Compared to direct injection into depleted oil/gas reservoirs or underground saline aquifers, in-situ carbon mineralization shows satisfactory characteristics in terms of safety and leakage risk by converting CO_2 into stable carbonate minerals (Cao et al., 2023; Kammerer et al., 2023; Snæbjörnsdóttir et al., 2020). However, considering this method as completely safe is not advisable. The reaction of CO_2 with basalt may lead to volume deformation, micro-crack propagation, deterioration of bonding, and initial damage accumulation in the formation material (Carrillo and Bourg, 2019). This is because the chemical composition of native minerals in basalt changes during this process, and there are differences in the physical properties compared to the generated secondary carbonate minerals. Results from CT scans demonstrate that the aforementioned fluid dynamics and geochemistry coupling behaviors can lead to particle displacement at the microscopic level. The changes in rock mechanical properties resulting from the dissolution of native minerals and the formation of secondary carbonate minerals in basalt after CO_2 injection should be further investigated (Ye et al., 2022). Moreover, changes in the mechanical properties of basalt formations may further activate and extend sealed faults, potentially leading to micro seismicity and felt earthquakes (Han et al., 2019).

Investigating the effects of fluid dynamics and geochemistry coupling behaviors on the static and dynamic mechanical properties of basalt is essential (Ye et al., 2022). Although some studies have explored the impact of fluid dynamics and geochemistry coupling behaviors on rock mechanical properties and permeability, these studies are evidently insufficient (Bakhshian and Sahimi, 2017). Ye et al. (2022) conducted experiments on CO_2-water-basalt reactions to analyze the effects of mineralization processes on the deterioration of basalt mechanical strength and increase in permeability, providing explanations through CT experiments, as shown in Fig. 10. In addition to changes in mechanical properties, alterations in basalt permeability are also worthy of attention (Spokas et al., 2019). Enhanced permeability may further increase the risk of leakage (Pollyea and Rimstidt, 2017; Wu et al., 2021). Additionally, porosity and permeability are positively correlated in most studies (Pollyea and Rimstidt, 2017). However, there are instances where permeability decreases with increasing porosity, as reported by Mangane et al. (2013), due to simultaneous pore volume expansion and pore throat blockage. The blocking of throats between pores can be attributed to fast mineral precipitation, non-uniform mechanical compression, etc. In studies related to in-situ mineralization, it is also worth investigating whether such an anomaly occurs in mafic rocks.

Furthermore, even though the reaction of CO_2 and rock has been proven to be rapid, complete consumption of carbon dioxide by geological formations still takes several years or longer, as evidenced by ongoing in-situ mineralization carbon storage projects such as Carbfix (McGrail et al., 2017; White et al., 2020). Long-term storage of CO_2 in formations may lead to issues such as surface uplift (Rahman et al., 2022).

6.1.2. Limitations on resolution and post-processing methods for imaging techniques

In recent years, imaging techniques have continuously made breakthroughs, providing support for understanding fluid dynamics and geochemistry coupling behaviors. These studies have significantly deepened the understanding of GCS systems (Rasaei and Sahimi, 2009). However, the limitations of resolution and post-processing methods of imaging techniques have been a persistent challenge in GCS research (Diwakar and Kumar, 2018).

The limitations of resolution are mainly evident at the nanoscale. When pore sizes are smaller than the resolution, imaging techniques become inadequate. For instance, in dense igneous rocks, fluid dynamics and geochemistry coupling behaviors are challenging to identify due to the small pore sizes (Sahimi and Hashemi, 2001). Cap rock rocks in GCS engineering are typically shales or tight sandstones, with numerous nanoscale pores. Scanning systems with nanoscale resolution help better elucidate the nanoscale etching behavior resulting from the interaction between cap rocks and carbon dioxide. However, most scanning technologies currently achieve micron-scale precision, and observing pores

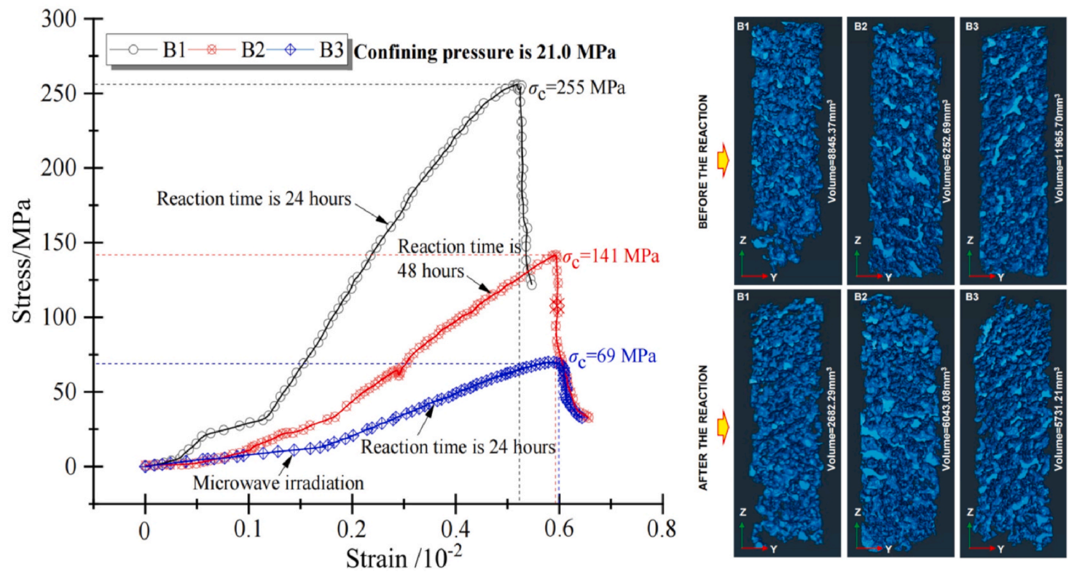

Fig. 10. The strength and pore structure changes of basalt after reaction with CO_2 (Ye et al., 2022).

at the nanoscale is costly (Sakdinawat and Attwood, 2010).

Post-processing plays a crucial role in accurately extracting image information. Images acquired from three-dimensional imaging techniques are vulnerable to noise and artifacts. Image noise, characterized by a "grainy" appearance resulting from random fluctuations in beam energy, significantly constrains the resolution of low-contrast scans. In the study by Noiriel et al. (2007), the liquid-rock interface evolved from a clear interface to a blurred, diffused interface as the liquid-rock reaction progressed (Fig. 11). The challenge of this work lies in balancing the conflicting demands between reducing noise and preserving sample information. Minimizing noise without losing image information is a challenging task (Bencsik and Ramanathan, 2001). Additionally, accurate registration and segmentation are crucial for subsequent analysis. Obtaining real-time images of samples during CO_2 injection or reaction with CO_2 is necessary and challenging (Diwakar and Kumar, 2018).

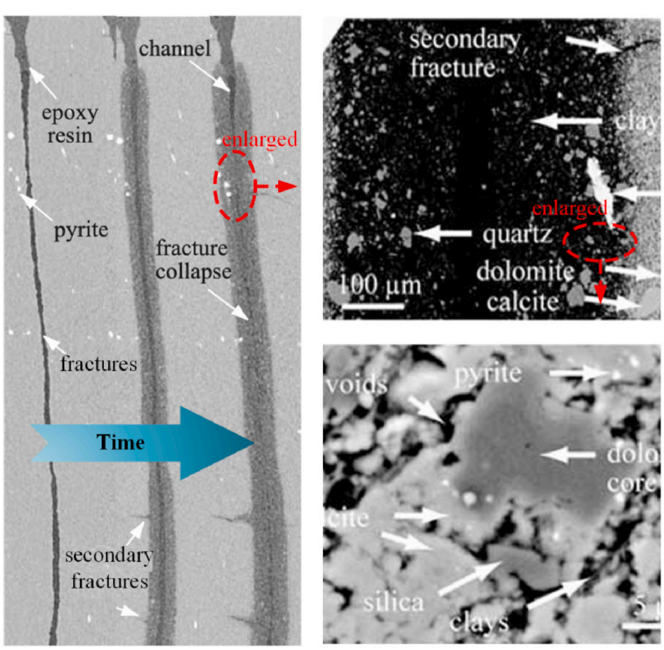

Fig. 11. A demonstration of the liquid-rock interface becoming blurred with the progress of the liquid-rock reaction (Noiriel et al., 2007).

Technologies that can simultaneously capture the three-dimensional structure of samples and their accurate chemical composition are difficult to achieve, increasing experimental perturbations (Sakdinawat and Attwood, 2010). In situ observation of large-sized samples is difficult and requires further development and research (Zhang et al., 2012).

Therefore, there is a need to develop experimental apparatus and image processing software. Although some algorithms and programs provide references for image denoising, there are still some issues present. The application of artificial intelligence may offer new perspectives for denoising in imaging techniques.

6.1.3. Influence of complex rock and gas compositions on the accuracy of RTM

In some cases, disparities between RTM and field observations may limit the application of RTM in engineering. Within this context, the issue of complex compositions must be considered, including the complex compositions in injected gases and reservoir materials (Spycher et al., 2019).

Geological materials (rocks) distributed on the earth also have complex mineral compositions, with varying types and ratios of minerals (Dessert et al., 2003; Oelkers and Cole, 2008). The presence of certain minerals significantly affects the action of CO_2. One typical example is the pH buffering capacity exhibited by some minerals. For example, rocks abundant in plagioclase and clay minerals can hinder a decrease in pH following CO_2 injection, thereby retarding the CO_2-rock reaction (Gherardi et al., 2007). The reduction in pH buffering capacity decreases the likelihood of CO_2 infiltrating the cap rock, offering an advantage for GCS systems. Current RTM involving rock-water-CO_2 interactions calculate accurately and with relatively low computational loads when considering single mineral compositions (Ennis-King and Paterson, 2007). Complex computational models pose challenges in terms of computational load.

The presence of impurity gases during CO_2 injection can affect the modeling and analysis process (Pearce et al., 2015). Generally, impurities present during CO_2 injection may include NO_2, SO_2, and N_2, among others (Knauss et al., 2003; Lei et al., 2021; Pearce et al., 2015). Even if the content of impurity gases is low, they can influence the dissolution and precipitation processes of CO_2, ultimately leading to discrepancies between model calculations and actual conditions. Spycher et al. (2019) discovered that when impurities such as SO_2 and NO_2 are introduced, they decompose into the aqueous phase, reducing pH values compared to CO_2 acidification alone (as shown in Fig. 12).

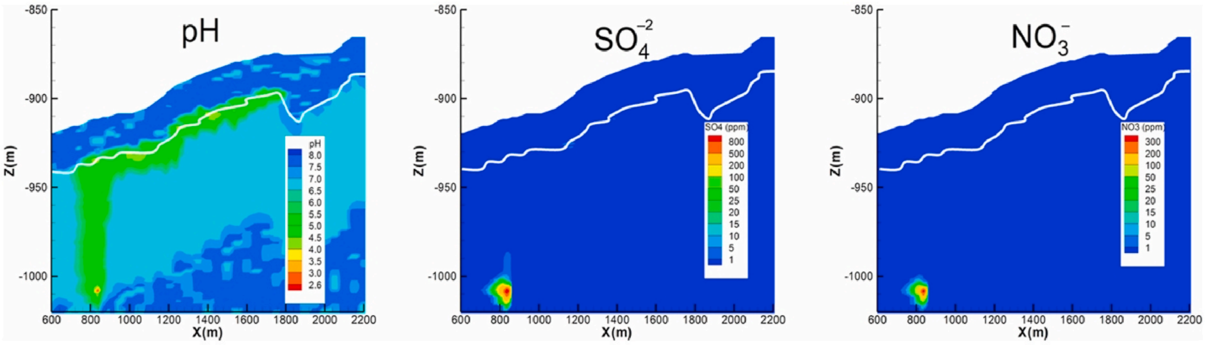

Fig. 12. Effects of NO_2 and SO_2 on formation pH and ion distribution (Spycher et al., 2019).

Additionally, the target reservoir may also contain other impurity gases such as H_2S (Jacquemet et al., 2011). In underground environments, sulfur-containing substances generate H_2S through the action of sulfate-reducing microbes and thermochemical reduction (Peng et al., 2022; Um et al., 2012). H_2S can also cause deviations in calculation results and exacerbate the risk of cement leakage in wells (Kutchko et al., 2015).

6.2. Future directions

6.2.1. Changes in rock properties induced by mineralization

As described in Sections 2 and 6.1.1, the changes in properties such as strength and permeability of the host rock with high reactivity(e.g., basalt) due to mineral sequestration based on fluid dynamics and geochemistry coupling is one of the future research directions. The mineral dissolution and precipitation enhancement processes induced by fluid dynamics and geochemistry coupling is a complex multi-field coupling process involving water, heat, mechanics, and chemistry (Blunt et al., 2013; Berkowitz et al., 2016). It is essential to consider all these factors comprehensively during the analysis.

Firstly, this process involves changes in the chemical composition of basic rocks. The dissolution of primary rock minerals and the formation of secondary carbonate minerals can significantly alter the mechanical properties of the rock (Zhang et al., 2012). The replacement of primary minerals by secondary minerals may lead to localized volume changes, causing microcracks to initiate and propagate, thereby disrupting the original stable structure of the rock and accumulating damage within it (Dashtian et al., 2019). This can result in a decrease in the rock's mechanical strength and an increase in permeability. Moreover, the mechanical properties of primary and secondary minerals often differ significantly. For example, the hardness and elastic modulus of olivine are nearly two times those of secondary carbonate minerals such as calcite, magnesite, and siderite (Marieni et al., 2020).

In addition, the mineral dissolution and precipitation enhancement process induced by fluid dynamics and geochemistry coupling involves changes in temperature and pressure (Bosworth, 1981; Feucht and Logan, 1990). The temperature at the depth of the formation is an uncertain parameter, varying significantly with depth and geographical location, ranging from a few degrees Celsius to several hundred degrees Celsius (Oelkers et al., 2015). Formation pressure also varies greatly depending on the type and depth of the overlying strata. Temperature and pressure can affect the CO_2 mineralization process, potentially impacting the post-reaction mechanical properties of the rock, and their influence on the mineralization reaction cannot be ignored (Feucht and Logan, 1990). The flow within the rock also affects its mechanical properties. Studies have shown that even soaking in pure water can affect the mechanical properties of rock to a certain extent (Brantley and Mellott, 2000; Jeschke and Dreybrodt, 2002). In GCS projects, the water carrying CO_2 entering the rock formation will lead to changes in the mechanical properties of the rock.

It is crucial to study the changes in the static and dynamic mechanical properties of basalt before and after mineralization reactions at microscopic, model, and engineering scales (Kumar et al., 2020). Future research should comprehensively consider factors such as the dissolution of primary minerals, the formation of secondary minerals, fluid interactions, and structural changes, and use water-heat-mechanics-chemistry coupling methods to establish models for the evolution of mechanical properties and permeability during the mineralization process of formation rocks.

6.2.2. High-resolution imaging technology development and post-processing algorithm optimization

The pore sizes of rocks in formations can range from nanometers to micrometers or millimeters. For instance, in GCS projects, shales often have pore sizes at the nanometer scale (Blunt et al., 2013). Currently, conventional micro-CT scanning systems cannot accurately identify the nanometer-scale pores in these dense rocks, thus necessitating further development of nanometer-scale CT scanning technology in the future. Thanks to the efforts of researchers, nanometer-scale CT scanning technology has been applied on a small scale in laboratories. However, besides the high cost, another issue with nanometer-scale CT scanning technology is the long scanning time, which hinders dynamic real-time imaging. The fluid dynamics and geochemistry coupling process in formations is a dynamic evolution process that continuously changes (Knauss et al., 2003). However, nanometer-scale CT cannot currently achieve real-time imaging at high resolution due to slow scanning speeds (Mayerich et al., 2022). In the future, synchrotron-based CT scanning systems should be further researched as they enable rapid observation at the nanometer scale.

As described in Section 6.1.2, post-processing methods are a challenging task in CT scanning. Some algorithms for reducing noise and artifacts already exist, such as the non-local means denoising algorithm (proposed by Buades et al., 2005) and the slice stacking method (Barrett and Keat, 2004). In future research, methods such as beam hardening filters, iterative reconstruction methods, and helical scanning trajectories (proposed by Cnudde and Boone, 2013) should be emphasized. Additionally, integrating various noise reduction techniques and packaging them into commercial post-processing software will be efficient. Providing intelligent noise reduction algorithms tailored to the conditions of different GCS projects should also be a future research direction.

6.2.3. Reactive transport modeling considering complex components

Analyzing the fluid dynamics and geochemistry coupling behavior involving complex components in GCS projects poses challenges to improving modeling efficiency and accuracy (Xu et al., 2011). The precise consideration of various complex components undoubtedly greatly increases the computational load of modeling and analysis. Areas with physical property discontinuities in the model often require denser grids and more refined calculations, leading to an exponential increase in computational load (Deng et al., 2022; Noiriel et al., 2007).

In future research, to address the multi-field, multi-scale challenges faced by many real geochemical systems, intelligent modeling capabilities are needed to optimize localization and select appropriate modeling methods. Current reactive transport codes primarily rely on serial computing, which is inefficient for handling complex problems (Xu et al., 2008). For issues involving complex formation materials and injected gas compositions, it is reasonable to consider developing parallel computing methods, which are expected to significantly reduce computation time (Ellis et al., 2013). The application of machine learning has been recognized as having great potential, especially with advancements in physics-informed algorithms, ensuring the ongoing development of high-fidelity process models involving complex components (Abgaze and Sharma, 2015b).

Analyzing the fluid dynamics and geochemistry coupling behavior involving complex compositions in GCS engineering presents a challenge in improving modeling efficiency and accuracy (Xu et al., 2011). To tackle the multifield and multiscale challenges encountered by many practical geochemical systems, intelligent modeling capabilities are necessary to optimize locality and select appropriate modeling methods (Mangane et al., 2013). The application of machine learning has been recognized to have tremendous potential, especially with advancements in physics-informed algorithms, ensuring the continued development of high-fidelity process-based models involving complex compositions.

7. Conclusion

The latest research progress in fluid dynamics and geochemistry coupling within GCS systems is reviewed in this article. A deeper understanding of fluid dynamics and geochemistry coupling behavior will improve the CO_2 storage efficiency of GCS projects, mitigate leakage, and reduce mechanical failure risks in GCS systems.

Although significant progress has been made in understanding fluid dynamics and geochemistry coupling behavior, several challenges still remain. This paper summarizes three main challenges: impact of CO_2 on the properties of basalt under the scenario of in-situ CO_2 mineralization in basalt, limitations on resolution and post-processing methods for imaging techniques, and selection of representative and precise parameters in upscaling numerical simulation. To overcome these challenges, further analysis on basalt properties in in-situ CO_2 mineralization, and integrity of wellbores in in-situ CO_2 mineralization is required. High-resolution imaging techniques applicable to in-situ experimental conditions needs to be developed to achieve in-situ high-precision imaging. Furthermore, reactive transport modeling considering complex compositions with reasonable requirement of computational resources is essential.

CRediT authorship contribution statement

Yue Yin: Writing – original draft, Methodology, Investigation, Conceptualization. **Liwei Zhang:** Writing – review & editing, Supervision, Software, Resources, Methodology, Conceptualization. **Hang Deng:** Methodology. **Yan Wang:** Resources. **Haibin Wang:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Abbreviations

CO_2	Carbon dioxide
H_2S	Hydrogen sulfide
GCS	Geologic CO_2 storage
RTM	Reactive transport modeling
MCM	Micro-continuum models
p	Porosity
VoF	Volume-of-Fluid
DNS	Direct Numerical Simulation
PNM	Pore network model
LS	Level-Set
REV	Representative Elementary Volume
K	Hydraulic conductivity
WT	Wavelet transform
CT	Computed Tomography
NMR	Nuclear Magnetic Resonance
SEM	Scanning Electron Microscopy
EDS	Energy Dispersive X-ray Spectroscopy

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